

Supplementary Notes for the Affine-Line Exponential-Period Theorem

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Abstract

These notes preserve constructions removed from the streamlined Paper II, *A Linear Kontsevich–Zagier Theorem for Compact Exponential Periods on the Affine Line*. They are not needed for its shortest proof. The most useful retained items are an explicit boundary presentation suited to computation, affine covariance, and an abstract spectral criterion behind direct-sum horizontal rigidity.

1 The explicit boundary presentation

Let $\Delta = \sum_{\alpha} d_{\alpha}[\alpha]$ be a finite degree-zero zero-cycle on $\mathbb{A}^1(\overline{\mathbb{Q}})$ with coefficients in $\overline{\mathbb{Q}}$. For $q, r \in \overline{\mathbb{Q}}[x]$, write

$$P(q, r, \Delta)$$

for a formal compact affine-line exponential period, and write $E(\beta)$ for the zero-dimensional exponential period e^{β} .

Definition 1.1 (Boundary presentation). Let $\mathcal{B}_{\mathbb{A}^1}^{\text{cpt}}$ be the $\overline{\mathbb{Q}}$ -vector space generated by these symbols, modulo linearity in r and Δ and the following relation. Let $Y/\overline{\mathbb{Q}}$ be a smooth connected affine curve, let $\pi \in \mathcal{O}(Y)$, and let η be a compact relative one-chain on $Y(\mathbb{C})$ with algebraic boundary

$$\partial\eta = \sum_p n_p[p].$$

Suppose that morphisms $\rho_{\nu} : Y \rightarrow \mathbb{A}^1$, polynomials $q_{\nu}, r_{\nu} \in \overline{\mathbb{Q}}[x]$, and coefficients $c_{\nu} \in \overline{\mathbb{Q}}$ satisfy

$$q_{\nu} \circ \rho_{\nu} = \pi$$

and

$$\sum_{\nu} c_{\nu} \rho_{\nu}^*(r_{\nu}(x) dx) = dG + G d\pi$$

for some $G \in \mathcal{O}(Y)$. Put $\Delta_{\nu} = \partial(\rho_{\nu*}\eta)$. Impose

$$\sum_{\nu} c_{\nu} P(q_{\nu}, r_{\nu}, \Delta_{\nu}) = \sum_p n_p G(p) E(\pi(p)). \quad (1)$$

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Relation (1) simultaneously encodes change of variables and twisted Stokes. It gives a well-defined period map

$$P(q, r, \Delta) \mapsto \int_{\Gamma} r(x) e^{q(x)} dx, \quad E(\beta) \mapsto e^{\beta},$$

where $\partial\Gamma = \Delta$.

There is a canonical surjection from this boundary presentation to the affine-line subspace of the curve-level formal period space used in the streamlined paper. Indeed, (1) follows there from functoriality, bilinearity, and twisted Stokes.

Proposition 1.2 (Equivalence of the two presentations). *Assuming the main theorem of Paper II, the boundary presentation and the affine-line subspace of the curve-level formal period space are canonically isomorphic.*

Proof. The canonical map is surjective. The proof of Paper II shows directly that a functional relation has vanishing inverse-branch densities. On the simultaneous Galois cover, each edge relation has the form (1) with $G = 0$; twisted reduction has the same form with $Y = \mathbb{A}^1$. Therefore the arithmetic transfer theorem proves that the period map on the boundary presentation is injective. Its canonical surjection to the curve-level subspace consequently has zero kernel. $\square$

The boundary presentation is longer and less standard, which is why it was removed from the main paper. It may nevertheless be useful for an algorithm that stores relations entirely in terms of affine-line boundary data and finite algebraic covers.

2 Affine covariance

Lemma 2.1 (Affine covariance). *Let $a \in \overline{\mathbb{Q}}^\times$, $b \in \overline{\mathbb{Q}}$, and put*

$$\tilde{q}(y) = q(ay + b), \quad \tilde{r}(y) = a r(ay + b), \quad \tilde{\Delta} = \sum_{\alpha} d_{\alpha} \left[\frac{\alpha - b}{a} \right].$$

Then

$$I_{q,r,\Delta}(z) = I_{\tilde{q},\tilde{r},\tilde{\Delta}}(z)$$

for all z , and the corresponding formal symbols are equal.

Proof. This is the change of variables $x = ay + b$. Formally, it is the functoriality relation for the automorphism $y \mapsto ay + b$ of $\mathbb{A}^1$. $\square$

This lemma remains useful for examples or normal forms, but it should not be used to normalize the polynomial in the proof: the unnormalized moment system already has the required residue spectrum, and normalization only introduces extra symbols and transformed cycles.

3 An abstract direct-sum rigidity criterion

The direct-sum argument depends only on spectral nonresonance, not on the particular origin of the moment systems.

Proposition 3.1 (Spectral criterion). *Let $C, D \in M_n(\mathbb{C})$, with C semisimple. Assume:*

- (i) *no negative integer is an eigenvalue of $\text{ad } D : X \mapsto [D, X]$;*

(ii) for every eigenvalue λ of C , no positive integer is an eigenvalue of

$$X \mapsto [\pi_\lambda D|_{\ker(C-\lambda I)}, X].$$

If $P \in M_n(\mathbb{C}(z))$ satisfies

$$P' = [C + D/z, P],$$

then P is constant and commutes with C, D .

Proof. A pole away from zero is excluded by comparing pole orders. At zero, the leading Laurent coefficient of a pole of order m would be a $-m$ -eigenvector of $\text{ad } D$, contrary to (i). Thus P is a polynomial. If its degree is $d > 0$, its leading coefficient commutes with C . Compressing the next coefficient equation to a C -eigenspace makes a nonzero leading block a d -eigenvector of the commutator in (ii), a contradiction. $\square$

A convenient sufficient condition is that the eigenvalues of D lie in an interval of length less than one, and that the eigenvalues of every compressed block lie in an interval of length less than one. The polynomial moment systems satisfy the stronger condition that all relevant spectra lie in $(-1, 0)$. This abstract form may be useful for other finite maps or other Fourier–Laplace connections.

4 Algebraic evaluation points

For $\xi \in \overline{\mathbb{Q}}^\times$,

$$I_{q,r,\Delta}(\xi z) = I_{\xi q,r,\Delta}(z).$$

Thus a finite relation involving separate algebraic arguments ξ_j is already a relation for the polynomials $\xi_j q_j$. Carrying the ξ_j as additional theorem parameters obscures this fact and is unnecessary.

There is one qualification. Twisted reduction of a general polynomial differential at the numerical value $z = \xi$ uses $R' + \xi q' R$, not $R' + q' R$. The clean procedure is therefore to absorb ξ into q first and then perform the canonical reduction for the new polynomial.

5 Terminology concordance

The streamlined papers use the vocabulary of the exponential-period, E -function, and polynomial-moment literature. The following replacements were made systematically:

Earlier draft	Streamlined terminology
polynomial phase	polynomial q or regular function
phase value	value of q
multiphase relation	relation involving several polynomials
phase moment	moment in q
phase density	inverse-branch density
common constellation	common tree representation
zero-chain	degree-zero zero-cycle
polynomial-line category	affine-line subspace
amplitude	polynomial differential or coefficient polynomial
evaluation parameter ξ	absorbed into q

The word *phase* remains appropriate in a stationary-phase, oscillatory, or Stokes-asymptotic argument. It is not needed as the generic name for the polynomial in these two papers.

6 Items deliberately not promoted to the main theorem

The explicit boundary presentation, affine normalization, and separate evaluation parameters are all valid. They were removed because they add formal overhead without strengthening the central statement. The abstract spectral criterion is more reusable, but including it in the main paper would interrupt the proof's one-dimensional narrative. These notes retain all four ideas for later algorithmic or geometric extensions.